SIMATS ENGINEERING
ASSESSMENT TOOL 3

COURSE NAME: DEEP LEARNING COURSE CODE: MLA0408

COURSE OUTCOME COVERED

CO2: Analyze Machine Learning and Deep Learning models by evaluating neural network architectures, representation learning, activation functions, unsupervised learning techniques, and their real-world applications. (L4)

CO Weightage: 15%

Assessment Tool Weightage: 20%

QUESTIONS: Total Marks: 20

1. Differentiate between Machine Learning and Deep Learning with respect to architecture, data requirements, feature extraction and applications. Give two real-world applications of each.

2. Explain the concept of Representation Learning. Discuss how the width and depth of a neural network influence its learning capability, computational complexity and model performance.

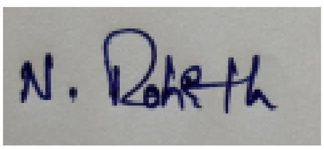
3. Compare ReLU, Leaky ReLU (LReLU) and ELU (Exponential Linear Unit) activation functions. Explain their mathematical behavior, advantages, disadvantages and suitable use cases.

4. Explain the concept of unsupervised learning in neural networks. Describe the working principle of Restricted Boltzmann Machines (RBMs) and Autoencoders and compare their objectives and applications.

Code of Conduct:

*I **NALIPI REDDY ROHITH/192425115(**Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the student:



RUBRICS FOR CALCULATION:

Criteria	Weigh ta ge	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Concept Understand in g	25%	Clearly explains all concepts with complete accuracy and depth	Minor gaps in understand in g	Basic understand in g with some errors	Incorrect or very limited understand in g
Content Coverage	25%	Covers all required topics comprehensi vel y	Covers most topics with minor omissions	Covers some topics only	Very few topics covered
Technical Accuracy	20%	All explanations are technically correct and precise	Minor technical errors	Some incorrect or vague explanations	Major technical errors
Integration of Concepts	15%	Effectively connects and relates concepts logically	Some connectio ns made with minor gaps	Limited connectio ns between concepts	No clear relationsh ip between concepts
Clarity & Presentation	15%	Well-structur ed, clear, and easy to understand	Mostly clear with minor issues	Somewhat unclear or poorly organized	Poorly structured and difficult to understand

COD-ATS

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MLA0408 - Deep learning

1. Introduction

Machine learning (ML) and Deep learning (DL) are two important branches of Artificial intelligence (AI). Both enable computer to learn from data and make predictions without being explicitly programmed. Machine learning uses statistical algorithms, while deep learning uses multi-layer Artificial Neural Network to automatically learn complex pattern from large amounts of data.

Difference Between ML and DL

feature	Machine learning	Deep learning
Architecture	uses traditional algorithms such as svm, logistic	uses (ANNs) with multiple hidden layers
Data requirement.	performs well with small or medium sized datasets.	Require large amounts of labeled data for better accuracy

feature extraction	feature are selected manually by domain experts	Automatically extracts important feature from raw
Accuracy	suitable for simple prediction tasks.	Achieves higher accuracy for complex tasks like image
interpretability	easier to understand and explain results.	more difficult to interpr because of deep network str

Advantages of ML

-> faster to train

-> easy to interpret

-> suitable for structured datasets.

Advantages of DL

-> learns complex patterns automatically.

-> eliminates manual features engineering.

-> performs well on image, audio, and text

Conclusion

ML is suitable for structured data and smaller d

while Deep learning is ideal for large datasets and com

applications

⑤ Introduction

Representation learning is a technique in Deep learn[ing] where a model automatically learns features from raw input data instead of relying on manually engineered features. It helps neural Networks discover hidden patterns and improve prediction accuracy.

Representing learning

Representing learning converts raw data into meaningful representations that are useful for learning tasks.

Example :-

-> In image recognition.

-> middle layers detect shapes.

-> final layer recognize complex object

Width of a Neural Network

Width refers to the number of neurons present in each hidden layer.

Advantages

-> learns more feature simultaneously.

-> improves prediction capability.

Disadvantages

Increases memory usage

-> Higher computational cost

-> may lead to overfitting if too many neurons are used

Depth of Neural Network

Depth refers to the number of hidden layers.

Advantages

-> Learns hierarchical and complex features.

-> Better representation learning

Disadvantages

-> Requires longer training time.

-> Needs powerful hardware

model Performance

-> Too few neurons or layers may cause underfitting

-> Too many neurons or layers may cause overfitting.

-> Proper architecture improves generalization.

Conclusion

Representation learning enables automatic feature extract

selecting an appropriate network width and depth helps

improve learning capability and over model Performance.

(3) Introduction

Activation functions introduce non-linearity into neur
networks, enabling them to solve complex problems.
ReLU, Leaky ReLU and ELU are among the most commo
used activation functions in Deep learning.

1. ReLU (Rectified linear unit)

formula:-

$$f(x) = \max(0, x)$$

Behaviour

* outputs zero for negative inputs.
* outputs the input itself for positive values.

Applications

→ image classification.
→ object Detection.

2. Leaky ReLU (LReLU)

formula:-

$$f(x) = \begin{cases} x, & x>0 \\ 0.01x, & x\leq 0 \end{cases}$$

Behaviour

Allows a small non-zero output for negative values.

Applications

-> Deep neural network.

-> GANS

3. ELU (Exponential linear unit)

formula :-

$$f(x) = \begin{cases} x, & x > 0 \\ \alpha(e^x - 1), & x \le 0 \end{cases}$$

Advantages

-> faster convergence

-> Better learning performance.

-> Reduces bias shift.

Disadvantages

-> computationally expensive

-> more complex implementation.

Conclusion

ReLU is simple and efficient, leaky ReLU solve the dead neuron problem, and ELU provides smoother learning and faster convergence.

(4) Introduction

unsupervised learning is a machine learning tech

where a neural network learns from unlabeled

instead of predicting target values, the model discov

hidden structures, similarities and patterns in the data

unsupervised learning

→ No class labels are provided.

→ the model automatically identifies meaningful patterns

→ used for clustering, feature learning.

Restricted Boltzmann machine (RBM)

An RBM is a two-layer neural network consisting of :

→ visible layer

→ Hidden layer

there are no connections b/w neurons within the

same layer.

Applications

→ collaborative filtering.

→ feature extraction.

Auto encoder

An encoder is a neural network consisting of:

- → encoder
- → Bottleneck (latent space)
- → Decoder.

Applications

- → image compression.
- → image denoising
- → dimensionality Reduction.

Advantages

RBM

- → learns hidden patterns.
- → effective for recommendation systems.

Autoencoder

- → Reduces dimensionality.
- → learns efficient feature representations.

Conclusion

ML and DL are powerful AI techniques used to solve a wide range of real-world problems, and the required level of accuracy.